PYQ · STATS I · SEP 2023

Find the IQR. Is there an outlier?

Use the IITM method, not your calculator. Answer on slide 3.

STATS I · QUARTILES

Your turn

Find the IQR of 14, 20, 40, 23, 18. Are there any outliers?

Comment your answer, then swipe.

ANSWER

IQR = 5, and 40 is an outlier

1. Sort: 14, 18, 20, 23, 40 (n = 5).
2. Q1: 0.25 × 5 = 1.25, round up to 2. The 2nd value is 18.
3. Q3: 0.75 × 5 = 3.75, round up to 4. The 4th value is 23.
4. IQR = 23 – 18 = 5. Fences: 18 – 7.5 = 10.5 and 23 + 7.5 = 30.5.
5. 40 is above 30.5, so it is an outlier.

TRAPS

Quartile mistakes

⚠ Round up even when the decimal is small: 1.25 becomes 2, not 1.

⚠ Calculators and Excel use different methods. Use the IITM steps.

⚠ Always sort the data first.

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